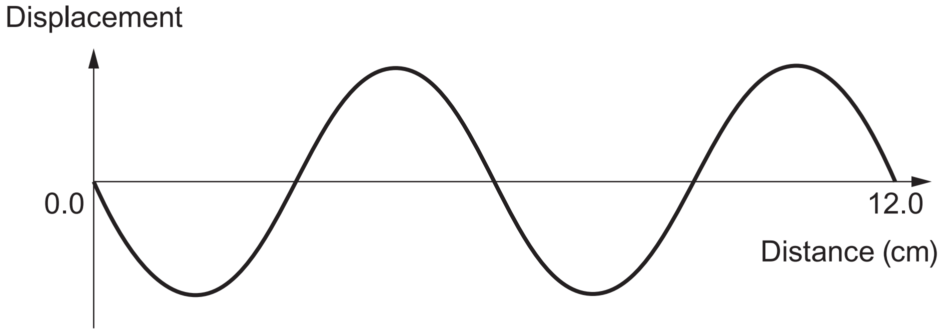
Examiner only

2. Microwaves travel through space at a speed of 3×10^8 m/s and are one part of the electromagnetic spectrum.

(a) The diagram represents microwaves produced at a certain frequency.



(i) Use the information in the diagram to calculate the wavelength of these microwaves. [1]

Wavelength = ... cm

(ii) Use your answer above and the equation:

wave speed = wavelength $\times$ frequency

to calculate the frequency of these microwaves. [3]

Frequency = ... Hz

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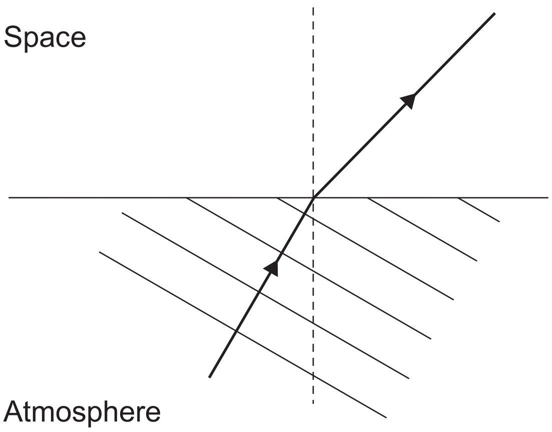
(b) Microwaves are refracted as they travel from the atmosphere into space.



Complete the diagram to show the refracted wavefronts in space. [3]

(c) Geostationary artificial satellites are used to send microwave signals around the world.

(i) Explain why a satellite in a geostationary orbit above the equator appears to stay in a fixed place above the surface of the Earth even though both the satellite and the Earth are constantly moving. [2]

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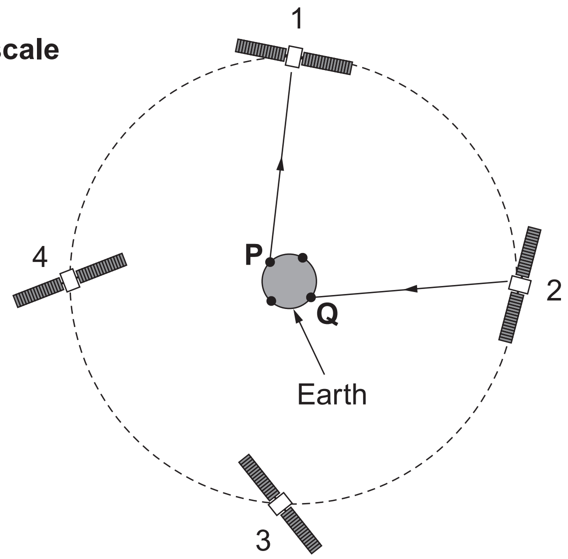
(ii) The diagram shows 4 satellites in geostationary orbit around the Earth.
A signal is to be sent from Earth station **P** to **Q** using the satellites 1 and 2.
The first and last parts of the path are shown.

Diagram not to scale



Use an equation from page 2 to calculate the time taken for the signal to travel from **P** to **Q**.
The height of a geostationary satellite above the Earth is 3.6×10^7 m. [3]
(Speed of light, $c = 3 \times 10^8$ m/s)

Time = ... s

(iii) It was suggested that if the signal had been sent in the opposite direction from **Q** to **P** via satellites **3** and **4** then it would have arrived quicker.
Explain whether you agree with this suggestion. [1]

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